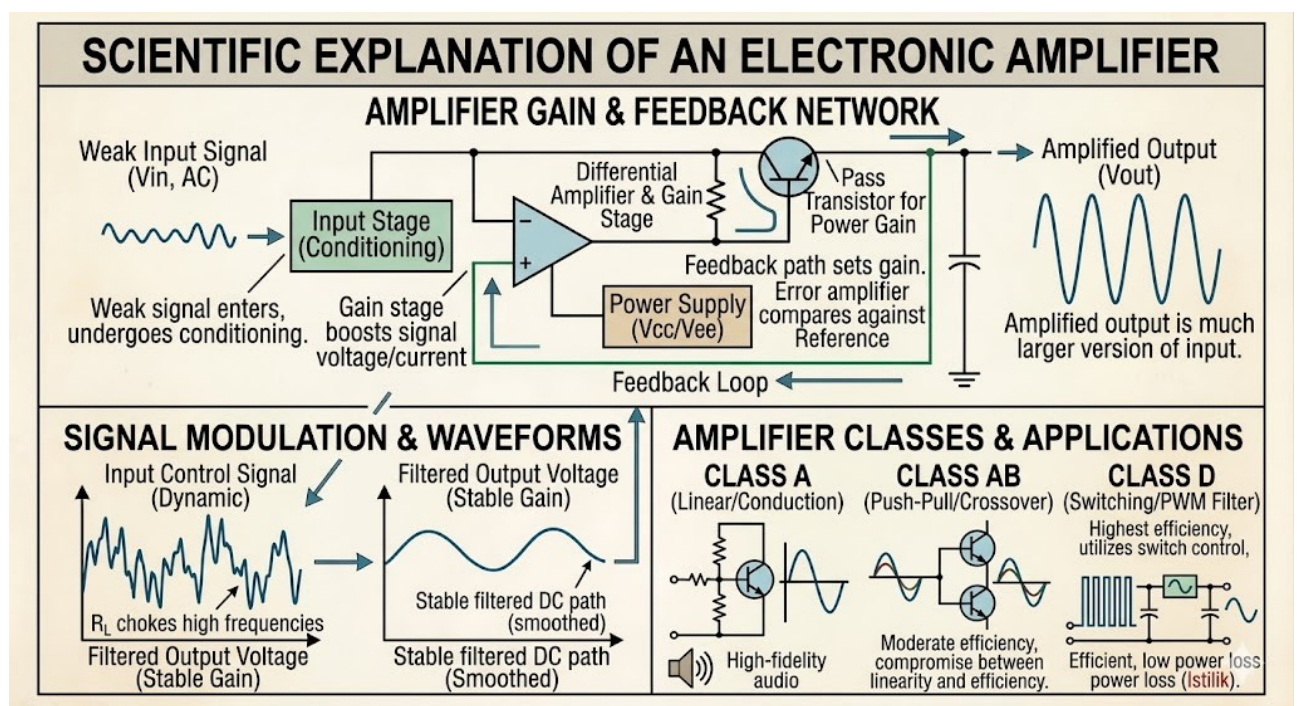


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## □ SECTION 1: AMPLIFIER GAIN & FEEDBACK NETWORK

An **electronic amplifier** is an active circuit that increases the power, voltage, or current of an electrical input signal. It uses energy from an external power supply to reshape the input signal into a much larger, higher-amplitude version at the output.

- **Weak Input Signal ( $V_{in}$ , AC):** A low-power, raw analog signal (such as an audio waveform or a sensor measurement) enters the amplifier circuit.
- **Input Stage (Conditioning):** The signal undergoes initial conditioning, which maximizes input impedance to prevent signal distortion or loading effects from the preceding stage.

- **Differential Amplifier & Gain Stage:** A high-gain core stage (often utilizing op-amps or pre-amplifier transistors) dramatically boosts the signal's voltage potential.
  - **Pass Transistor for Power Gain:** To deliver actual power to a heavy load (like a speaker or an antenna), a robust output transistor stage modulates large amounts of current from the primary **Power Supply** ( $V_{cc}/V_{ee}$ ).
  - **Closed Feedback Loop:** A portion of the amplified output is sent back to the differential stage. This negative feedback loop continuously stabilizes the system, dampens distortion, and fixes the overall system voltage gain to a highly predictable level.
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## III SECTION 2: SIGNAL MODULATION & WAVEFORMS

Amplification requires taking a low-energy control waveform and translating it into a heavy-duty power path without altering the signal's core information.

### 1. Dynamic Input Control Signal

- **Behavior:** The raw input waveform exhibits rapid fluctuations, noise, and microvolt-level variations. If connected directly to a heavy load, the signal would collapse entirely due to impedance mismatches.

### 2. Filtered Output Voltage (Stable Gain)

- **Behavior:** By drawing continuous power from the stable DC rails of the power supply, the amplifier replicates the input shape perfectly while scaling its magnitude up. Output filtering capacitors smooth out any residual high-frequency switching or power supply noise, leaving a clean, high-fidelity amplified AC waveform superimposed over a stable DC path.
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## ✂ SECTION 3: AMPLIFIER CLASSES & APPLICATIONS

Electronic amplifiers are categorized into different "Classes" based on their conduction angle, linearity, and power efficiency:

### 1. Class A (Linear/Conduction)

- **Operation:** The internal transistors conduct current through the entire  $360^\circ$  cycle of the input wave. The device is always completely "ON" and biased in its active region.

- **Characteristics:** It provides the highest linearity and ultra-low distortion (**High-fidelity audio**). However, it is highly inefficient (typically around 25%), wasting most of its supply energy as continuous heat (**istilik**).

## 2. Class AB (Push-Pull/Crossover)

- **Operation:** A design compromise that utilizes a pair of transistors in a "push-pull" configuration. One transistor amplifies the positive half of the wave, while the other handles the negative half. Both conduct slightly more than  $180^\circ$  to eliminate dead-zone distortion at the zero-crossing point.
- **Characteristics:** Offers a sweet spot with moderate-to-high efficiency and strong linearity, making it the classic standard for analog audio power amplifiers.

## 3. Class D (Switching/PWM Filter)

- **Operation:** Completely non-linear. The input signal is converted into a high-frequency digital Pulse Width Modulation (PWM) square wave. The output transistors act strictly as rapid electronic switches, flicking between fully ON (Saturation) and fully OFF (Cut-off).
  - **Characteristics:** Because the transistors spend virtually no time in the resistive active region, power loss as heat (**istilik**) is minimized, pushing efficiency beyond 90%. A low-pass LC filter at the output smooths the PWM back into a high-power analog wave.
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## 5. CRITICAL AMPLIFIER PARAMETERS

When evaluating or designing an electronic amplifier in software like Proteus or testing it on a breadboard, engineers analyze these specific performance boundaries:

- **Bandwidth (BW) & Frequency Response:** An amplifier cannot boost all frequencies equally. The bandwidth is the specific range of frequencies (from the lowest to the highest) where the amplifier operates efficiently with a power drop of less than  $3\text{ dB}$  (the half-power points).
- **Slew Rate (SR):** This represents the maximum speed at which the amplifier's output voltage can change in response to a sudden step-change at the input. Measured in Volts per microsecond ( $\text{V}/\mu\text{s}$ ), a low slew rate causes high-frequency signals to distort into triangular waves.

- **Total Harmonic Distortion (THD):** A percentage metric that quantifies how much unwanted harmonic frequency noise the amplifier adds to the original signal. For high-fidelity audio amplifiers, the THD must be kept as close to 0% as possible.
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## 6. UNDERSTANDING DISTORTION TYPES

If an amplifier is driven beyond its operational limits or improperly biased, the output waveform will experience severe distortion:

- **Clipping (Overdrive Distortion):** Occurs when the input signal is too large, causing the requested output voltage to exceed the power supply rails ( $V_{cc}/V_{ee}$ ). The tops and bottoms of the sine wave are physically "clipped" flat, turning the signal into a distorted, square-like wave.
- **Crossover Distortion:** Specifically affects Class B and improperly biased Class AB push-pull amplifiers. It happens during the brief moment when the signal transitions through  $0\text{V}$  and one transistor turns off before the other fully turns on, leaving a flattened "dead zone" in the middle of the waveform.